Programme

The second edition's programme is not announced yet. Below are the notes on laptops and setup, and the first edition's programme for reference.

Laptops and setup

Bring a laptop. Participants run the course's notebooks on their own computers, from each notebook's page on this site.

The setup instructions come with Before the school, the first topic of the second edition's course, which is not published yet.

First edition, 2025

1–3 October 2025 at AgroParisTech. Times are Paris time.

Day 1 Wednesday 1 October

09:00–09:30	Welcome, introduction and coffee
09:30 10:15	**Controlling complex microbiomes with computational intelligence: physical and virtual prototypes** Daniel Garza · Lecture
10:30 12:30	**Statistical mapping of structure-function landscapes in microbiomes** Djordje Bajić · Lecture
12:30–13:30	Lunch
13:30 15:30	**Computational design of synthetic microbial communities** Alex Fedorec · Lecture
15:45 17:45	**Integrating human gut microbiome data at scale: from individuals to multi-layered features in clinical and nutritional studies** Julien Tap · Lecture

Day 2 Thursday 2 October

09:30 10:15	**Bacterial reservoir computing** Jean-Loup Faulon · Lecture
10:30 12:30	**Simulation and control of high-dimensional dynamical systems using artificial neural networks** Lucas Böttcher · Lecture
12:30–13:30	Lunch
13:30 15:30	**Metabolic network reconstruction and modelling – from the individual to the community** Clémence Frioux · Lecture
15:45 17:45	**Sampling the solution space of metabolic networks** Haris Zafeiropoulos · Lecture

Day 3 Friday 3 October

09:30
12:30

Mapping the microverse and modelling its drivers

Bas Dutilh · Lecture

School of Artificial Intelligence Applied to Microbiomes

Second edition · Dates to be announced

Contact

schoolaisaclay@gmail.com

First edition, 2025

ai-microbiome-school.onrender.com